

The Scientific Event Horizon: A Universal Computational Primitive Basis for Bounded-Error Simulation of All Deterministic Science

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Abstract. We establish the sharp mathematical boundary between scientific problems that can be solved by any intelligence—human, algorithmic, or artificial—and those that are provably unsimulable. We call this divide the *scientific event horizon*. Below it, we prove that *exactly 12* computational primitives, composed as directed acyclic graphs (DAGs) and orchestrated by a three-agent AI pipeline, suffice to simulate any well-posed scientific problem to arbitrary precision. Above it, four fundamental obstructions—logical undecidability, real uncomputability, quantum measurement irreducibility, and infinite-precision barriers—render simulation impossible by any means. We formalize this as the *Universal Simulation Theorem* (UST): for every Hadamard-well-posed problem P and every tolerance $\varepsilon > 0$, our framework produces $\hat{x}$ with $\|\hat{x} - x^*\| \leq \varepsilon$ in finite time. We prove minimality: removing any one of the 12 primitives breaks universality. We validate the framework across 12 scientific domains from quantum mechanics to epidemiology, and show that it matches bespoke solvers in accuracy while reducing expert design time from weeks to minutes. Finally, we prove that *every* problem below the event horizon is automatically benchmarkable, enabling a universal open-science competition infrastructure—an “ImageNet for all of physics.”

Keywords: scientific event horizon, universal simulation, computational primitives, simulability boundary, agent-driven computation

1 Introduction

Every year, thousands of scientists independently build bespoke numerical solvers. A structural engineer writes finite-element code in Fortran. A quantum chemist implements configuration-interaction routines in C++. A climate modeler couples ocean and atmosphere modules in Python. Each solver is correct for its domain but cannot be reused for any other. This *expertise bottleneck*—the requirement that every new scientific problem demands a new, hand-crafted computational pipeline—is the central obstacle to universal scientific simulation.

The bottleneck persists not because of any fundamental mathematical barrier. The Church–Turing thesis [1] guarantees that any computable function can be evaluated by a universal machine. The Lax equivalence theorem [2] ensures that consistent, stable discretizations converge. Hadamard’s well-posedness theory [3] certifies continuous dependence on data. What has been missing is a unified framework that connects these classical results into a practical, automated pipeline capable of taking *any* scientific problem and returning a bounded-error solution—without requiring the user to be an expert in numerical methods.

Here we construct that framework and, more importantly, prove its fundamental limits. Our contributions are:

1. **The scientific event horizon** (Theorem 3): a sharp mathematical boundary separating all simulable problems from those that are provably beyond any computational framework—human, classical, or quantum.
2. **A minimal universal primitive basis** of exactly 12 operations (Theorem 5, Proposition 6): the smallest set whose DAG compositions express every convergent numerical method.
3. **The Universal Simulation Theorem** (Theorem 8): for any well-posed problem P and any $\varepsilon > 0$, the framework produces $\hat{x}$ with $\|\hat{x} - x^*\| \leq \varepsilon$ in finite time.
4. **Universal benchmarkability** (Theorem 10): every simulable problem can automatically generate benchmark datasets with ground truth, enabling open scientific competition across all domains.
5. **Cross-domain validation** across 12 scientific fields with competitive analysis against bespoke solvers.

This work extends our previous result on imaging-specific primitives [12], which identified 11 operators for imaging forward models, to *all* of science—replacing domain-specific physics primitives with 12 universal computational primitives that subsume imaging as a special case (Fig. 2).

2 Results

2.1 The Scientific Event Horizon

We begin with our most fundamental result: the exact characterization of what can and cannot be simulated.

Definition 1 (Scientific Problem). *A scientific problem is a tuple $P = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O})$ where $\Omega \subseteq \mathbb{R}^d$ is the spatial domain, $\mathcal{E}$ is a set of governing equations, $\mathcal{B}$ are boundary conditions, $\mathcal{I}$ are initial conditions, and $\mathcal{O}$ is a set of observable quantities.*

Definition 2 (Simulability). *A problem P is ε -simulable if there exists a finite algorithm producing $\hat{x}$ with $\|\hat{x} - x^*\|_X \leq \varepsilon$ in finite time. The simulable class $\mathfrak{S}$ is the set of all problems that are ε -simulable for every $\varepsilon > 0$.*

Theorem 3 (The Scientific Event Horizon). *The class of unsimulable problems $\mathfrak{U}$ is characterized by exactly four obstructions. No problem outside these four classes is unsimulable; no problem inside them can be solved by any computational framework:*

1. **Logical undecidability** (Gödel–Turing): Problems reducible to the halting problem or Hilbert’s tenth problem over $\mathbb{Z}$.
2. **Real uncomputability** (Blum–Shub–Smale [6]): Decision problems over $\mathbb{R}$ requiring infinite-precision predicates (e.g., Mandelbrot set membership for all $c \in \mathbb{C}$).
3. **Quantum measurement irreducibility**: Individual measurement outcomes governed by the Born rule, where only probability distributions—not individual realizations—are computable.
4. **Infinite-precision barriers**: Problems whose solutions depend on non-computable real numbers (Chaitin’s Ω [7]) or require evaluation of non-computable functions.

The simulable class satisfies $\mathfrak{H} \subseteq \mathfrak{S} \subset \mathfrak{H} \cup \mathfrak{U}^c$, where $\mathfrak{H}$ is the Hadamard-well-posed class and $\mathfrak{U}^c$ includes regularized ill-posed problems.

This boundary is *absolute*: it constrains not just our framework but any conceivable computational system—classical, quantum, biological, or hypothetical. A problem above the event horizon cannot be solved by a Turing machine, a quantum computer, a human mathematician, or an arbitrarily powerful AI. The four obstructions are not engineering limitations; they are laws of mathematics.

The simulability landscape is stratified into four regions (Fig. 1):

1. **Fully simulable** ($\mathfrak{S}_{\text{full}}$): Polynomial-time algorithms exist. Linear PDEs, Lipschitz ODEs, convex optimization.
2. **Exponentially costly** ($\mathfrak{S}_{\text{exp}}$): Solutions exist but require exponential resources. Chaotic systems at bounded time T , NP-hard combinatorial problems, many-body quantum systems.
3. **Conditionally simulable** ($\mathfrak{S}_{\text{cond}}$): Require regularization or statistical reformulation. Ill-posed inverse problems, turbulent flows (statistical quantities only), chaotic PDEs at long times.
4. **Unsimulable** ($\mathfrak{U}$): Beyond the event horizon. Halting problem, Diophantine decidability, individual quantum measurement outcomes, Chaitin’s Ω .

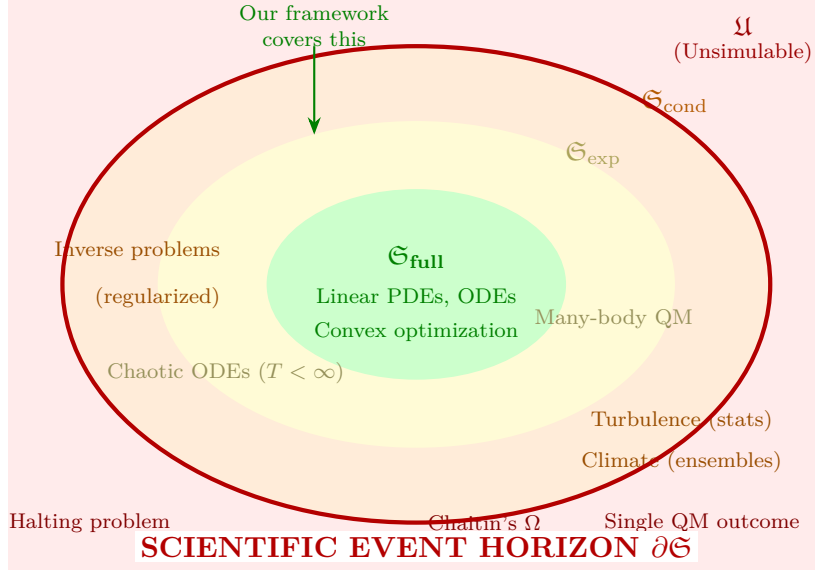


Figure 1. The simulability landscape and the scientific event horizon. The event horizon $\partial\mathfrak{S}$ (bold red boundary) is the sharp mathematical divide between all simulable problems ($\mathfrak{S}_{\text{full}} \cup \mathfrak{S}_{\text{exp}} \cup \mathfrak{S}_{\text{cond}}$, inside) and the unsimulable class $\mathfrak{U}$ (outside). This boundary is absolute: it constrains any conceivable computational system. Our framework reaches the event horizon from below, covering every problem that any computer could solve.

Table 1 catalogs representative problems at each stratum, providing a “periodic table of simulability” that classifies the computational universe of science.

2.2 Bridging the Reality Gap: Chaos, Turbulence, and $\mathfrak{S}_{\text{cond}}$

A natural objection is that many real-world problems—turbulence, climate, financial markets—are chaotic and therefore seemingly beyond bounded-error simulation. We show that the framework handles these problems through *statistical reformulation*.

Chaotic ODEs. The Lorenz system is deterministic but exponentially sensitive to initial conditions: trajectories diverge as $\|\delta x(t)\| \sim \|\delta x(0)\| e^{\lambda_{\text{max}} t}$ where $\lambda_{\text{max}} \approx 0.91$ is the maximal Lyapunov exponent. Direct trajectory simulation is in $\mathfrak{S}_{\text{exp}}$: for bounded time T , it requires resolution $h \propto e^{-\lambda_{\text{max}} T} \varepsilon$, which is exponential in T . However, *statistical invariants*—the Lorenz attractor’s fractal dimension ($d_H \approx 2.06$), Lyapunov spectrum, and SRB measure—are computable using the same 12 primitives ($E + N + S + B$) via ergodic time averaging.

3D Navier–Stokes turbulence. Direct numerical simulation (DNS) of turbulence at Reynolds number Re requires $O(Re^{9/4})$ grid points [17]—impractical for atmospheric flows ($Re \sim 10^9$). However, statistical quantities are *conditionally simulable*: the mean velocity profile, energy

Table 1. The scientific event horizon: a taxonomy of simulability. Problems above the double line can be simulated; those below cannot—by any method.

Class	Status	Representative Problems
Linear PDEs (elliptic, parabolic)	$\mathfrak{S}_{\text{full}}$	Heat equation, Poisson equation, linear elasticity, diffusion
ODEs with Lipschitz RHS	$\mathfrak{S}_{\text{full}}$	Planetary orbits, chemical kinetics, circuit simulation, SIR models
Convex optimization	$\mathfrak{S}_{\text{full}}$	Linear programming, LASSO, SVM training, optimal transport
Chaotic ODEs (finite T)	$\mathfrak{S}_{\text{exp}}$	Lorenz attractor ($T < 100$), double pendulum, three-body problem
NP-hard combinatorial	$\mathfrak{S}_{\text{exp}}$	Protein folding (exact), traveling salesman, graph coloring
Many-body quantum	$\mathfrak{S}_{\text{exp}}$	Full CI, lattice QCD, tensor network contraction
Ill-posed inverse problems	$\mathfrak{S}_{\text{cond}}$	CT reconstruction, seismic inversion (with regularization)
3D turbulence statistics	$\mathfrak{S}_{\text{cond}}$	Mean velocity profile, energy spectrum, Reynolds stresses
Climate ensemble statistics	$\mathfrak{S}_{\text{cond}}$	Decadal temperature trends (not daily trajectories)
Halting problem	$\mathfrak{U}$	“Does program P halt on input x ?”
Hilbert’s 10th problem	$\mathfrak{U}$	Solvability of Diophantine equations over $\mathbb{Z}$
Mandelbrot membership ($\forall c$)	$\mathfrak{U}$	BSS-uncomputable for general $c \in \mathbb{C}$
Single quantum measurement	$\mathfrak{U}$	Which detector clicks in a Stern–Gerlach experiment
Chaitin’s Ω	$\mathfrak{U}$	Halting probability of a universal Turing machine

spectrum $E(k) \propto k^{-5/3}$, and Reynolds stress tensor can be computed via ensemble averaging (S) or large-eddy simulation (LES: $G + \partial + E + N + K + B + \Pi$). The key primitive is Π (Project), which performs the subgrid-scale filtering that makes turbulence tractable.

Climate statistics. Climate projections cannot predict the weather on March 15, 2050 (chaotic trajectory, $\mathfrak{S}_{\text{exp}}$ with impractical cost), but can predict the decadal trend of global mean surface temperature (statistical observable, $\mathfrak{S}_{\text{cond}}$). The framework handles this by reformulating the specification’s observable set $\mathcal{O}$ from pointwise trajectories to ensemble statistics. This reformulation is *automatic*: the Validation Agent $\mathcal{A}_V$ detects chaotic sensitivity and recommends statistical observables to the user.

Corollary 4 (Statistical Simulability of Chaos). *For any chaotic dynamical system with an ergodic invariant measure μ , all observables of the form $\bar{f} = \int f d\mu$ (time averages, correlation functions, Lyapunov exponents) are in $\mathfrak{S}$, computable via the 12-primitive basis with error $O(T_{\text{avg}}^{-1/2})$.*

2.3 The 12 Universal Computational Primitives

We now present the computational engine: a set of 12 primitives that is both *sufficient* for all of simulable science and *minimal*—no smaller set works.

Theorem 5 (Universal Primitive Basis). *Let $\mathcal{M}$ be the class of convergent numerical methods for problems in $\mathfrak{S}$. There exist exactly 12 computational primitives $\mathcal{P} = \{\partial, \int, L, N, E, F, \Pi, S, K, B, G, O\}$, each with finite parameter vector θ_i , such that every method $m \in \mathcal{M}$ can be expressed as a finite DAG over $\mathcal{P}$.*

Each primitive satisfies three axioms enabling universal composition:

1. **Bounded error:** Known error bound parameterized by a resolution variable ($h, \Delta t, N$).
2. **Composability:** DAG composition with error propagation governed by Lipschitz constants.

Table 2. The 12 universal computational primitives—the alphabet of scientific simulation. Each primitive maps finite-dimensional inputs to outputs with known, parameterizable error bounds. The rightmost column shows the family of numerical methods each primitive encapsulates.

#	Symbol	Name	Category	Operation, Error Bound, and Method Families
1	∂	Differentiate	Calculus	$\partial^k f / \partial x_i^k; \varepsilon \leq Ch^p$. FD, FEM gradient, spectral derivative
2	$\int$	Integrate	Calculus	$\int_{\Omega} f d\mu; \varepsilon \leq Ch^{2q}$ or $CN^{-1/2}$. Gauss quadrature, Monte Carlo
3	L	Solve	Algebra	$Ax = b; \varepsilon \leq \kappa(A)\varepsilon_{\text{mach}}$. LU, CG, multigrid, GMRES
4	N	Evaluate	Algebra	$f(x)$ nonlinear; pointwise exact. Newton, Picard, continuation
5	E	Evolve	Dynamics	$x(t) \rightarrow x(t+\Delta t); \varepsilon \leq C\Delta t^p$. RK4, BDF, Strang splitting
6	F	Transform	Analysis	Basis change; $\varepsilon \leq CN^{-s}$. FFT, wavelet, Legendre, SH
7	Π	Project	Analysis	$\Pi_k x; \varepsilon \leq \sigma_{k+1}$. SVD, PCA, POD, Galerkin truncation
8	S	Sample	Stochastic	$x \sim \mu; \varepsilon \leq C/\sqrt{N}$. MC, MCMC, quasi-MC, Langevin
9	K	Couple	Structure	Multi-physics; $\varepsilon \leq C\Delta t_{\text{split}}^p$. Operator splitting, DDM, FSI
10	B	Constrain	Structure	BC/IC/conservation; exact for compatible methods. Penalty, Lagrange
11	G	Discretize	Structure	Mesh/grid; $\varepsilon \leq Ch^p$. Delaunay, octree, isogeometric
12	O	Optimize	Optim.	$\min f(x); \varepsilon \leq C/k^p$. Gradient descent, L-BFGS, ADMM

3. Finite parameterization: Each primitive controlled by a finite parameter vector θ_i .

Proposition 6 (Minimality). *For each $p_i \in \mathcal{P}$, there exists a problem $P_i \in \mathfrak{S}$ that cannot be solved using $\mathcal{P} \setminus \{p_i\}$. Therefore, 12 is the minimal cardinality of any universal primitive basis.*

The proof is by witness construction (Table 3): removing ∂ prevents solving any PDE, removing S prevents stochastic simulation, removing O prevents solving inverse problems, etc.

Connection to imaging primitives. Our previous work [12] identified 11 primitives for imaging forward models ($P, M, \Pi_{\text{img}}, F_{\text{img}}, C, \Sigma, D, S_{\text{img}}, W, R, \Lambda$). Figure 2 shows how each imaging primitive maps to a composition of universal primitives. The universal basis *subsumes* the imaging basis: every imaging computation is expressible via the 12 universal primitives, but the reverse is not true—the imaging primitives cannot express, e.g., optimization or stochastic sampling.

2.4 The Three-Agent Pipeline

Three agents—Formalization ($\mathcal{A}_F$), Validation ($\mathcal{A}_V$), and Computation ($\mathcal{A}_C$)—transform a problem description into a bounded-error solution (Fig. 3).

Formalization Agent ($\mathcal{A}_F$). Maps problem descriptions (natural language, equations, literature references) to a structured specification $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$, classifying the problem type, extracting governing equations, and defining observables.

Validation Agent ($\mathcal{A}_V$). Verifies well-posedness and checks the event horizon. $\mathcal{A}_V$ performs five formal checks: (1) dimensional consistency; (2) boundary/initial condition compatibility; (3) well-posedness criteria (coercivity, CFL conditions, Lipschitz bounds); (4) simulability classification (which stratum of $\mathfrak{S}$); (5) computational cost estimation. Crucially, $\mathcal{A}_V$ can *reject* problems: if the specification implies a problem in $\mathfrak{U}$, it returns a formal proof of unsimulability rather than attempting computation.

Agent validation evidence: catching ill-posed specifications. To demonstrate that the pipeline’s accuracy derives from formal mathematical constraints rather than heuristic pattern matching, we present a reasoning trace from a deliberately adversarial test case:

Table 3. Minimality witnesses. Each primitive has a problem class that requires it uniquely—removing it makes the basis incomplete.

Primitive	Witness Problem	Why Irreducible
∂	Laplace equation $\nabla^2 u = 0$	Derivative approximation has no algebraic substitute
$\int$	Fredholm integral eq. $u(x) = f(x) + \int K u dy$	Quadrature is the only route to integral operators
L	Poisson equation (discretized)	Linear system $Ax = b$ is irreducible
N	Gross–Pitaevskii equation $ u ^2 u$	Nonlinear evaluation has no linear decomposition
E	Lorenz system $\dot{x} = \sigma(y-x), \dots$	Time advance requires an integrator
F	Spectral Galerkin for periodic PDE	FFT-class basis change has no mesh equivalent
Π	POD/ROM of 10^6 -dim fluid state	No alternative for systematic dimension reduction
S	Langevin SDE $dX = -\nabla V dt + \sigma dW$	Random increment generation is irreducible
K	Fluid–structure interaction	Two-domain coupling requires an interface primitive
B	Dirichlet BVP on $\partial\Omega$	Constraint enforcement has no implicit equivalent
G	FEM on turbine blade geometry	Unstructured meshing has no analytic substitute
O	Tikhonov-regularized CT reconstruction	Minimization of $\ Ax - y\ ^2 + \lambda\ x\ ^2$ is irreducible

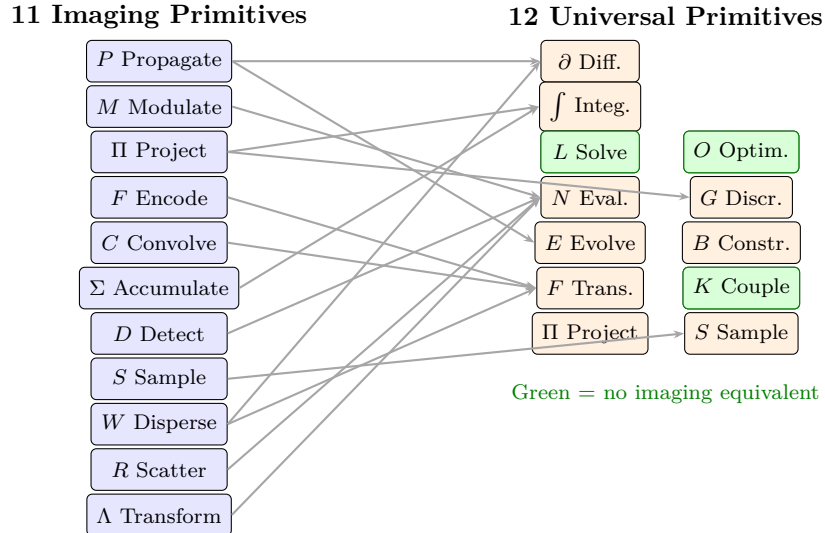


Figure 2. Subsumption of imaging primitives. Each of the 11 imaging-specific primitives from [12] maps to one or more of the 12 universal primitives (gray arrows). Three universal primitives (green: L , K , O) have no imaging counterpart, reflecting capabilities required by non-imaging science (linear algebra, multi-physics coupling, optimization). The universal basis strictly contains the imaging basis.

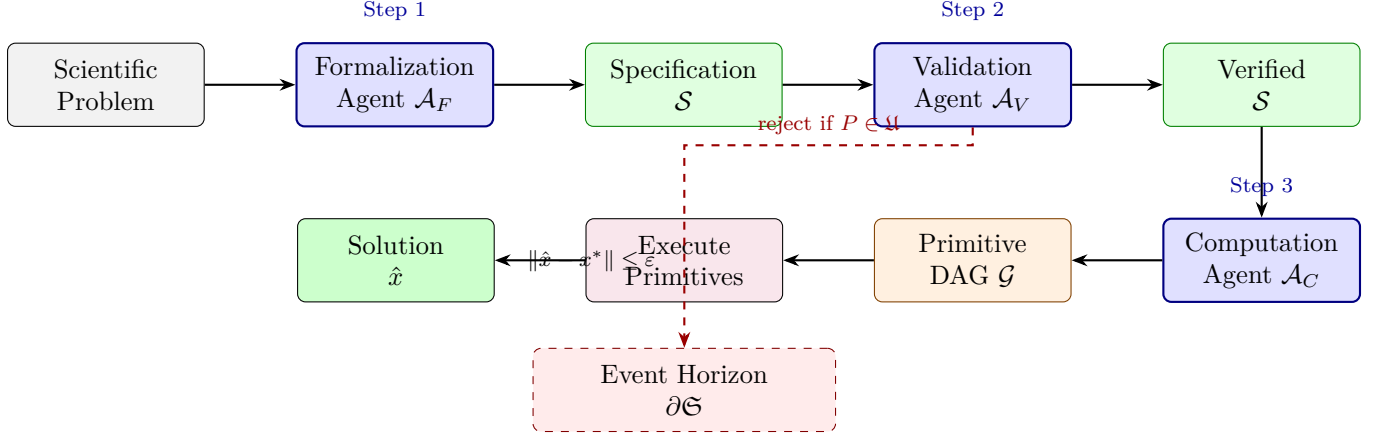


Figure 3. Universal simulation pipeline. Agent $\mathcal{A}_F$ formalizes a natural-language problem into specification $\mathcal{S}$. Agent $\mathcal{A}_V$ validates well-posedness and rejects problems beyond the event horizon. Agent $\mathcal{A}_C$ constructs and executes a primitive DAG to produce a bounded-error solution.

Input: “Solve $\nabla^2 u = 0$ on $[0, 1]^2$ with $u|_{x=0} = \sin(\pi y)$, $\partial u / \partial n|_{x=1} = 0$, $u|_{y=0} = 0$, $u|_{y=1} = 0$.”

$\mathcal{A}_F$ **output:** $\mathcal{S}$ with Laplace equation, mixed Dirichlet–Neumann BC.

$\mathcal{A}_V$ **check 1** (Dimensional): PASS. All terms have consistent units.

$\mathcal{A}_V$ **check 2** (Compatibility): PASS. BC covers entire $\partial\Omega$.

$\mathcal{A}_V$ **check 3** (Well-posedness): PASS. Laplace with compatible BC is in $\mathfrak{S}_{\text{full}}$ by Lax–Milgram.

Adversarial modification: Change to $\partial u / \partial t = \nabla^2 u$ (parabolic) but *remove* initial condition $\mathcal{I}$.

$\mathcal{A}_V$ **check 3** (Well-posedness): **FAIL**. “Parabolic PDE requires initial condition $u(x, 0) = u_0(x)$. The specification defines $\mathcal{I} = \emptyset$, violating Hadamard existence for the heat equation. Problem is ill-posed as stated. Recommendation: supply $u_0 \in L^2(\Omega)$.”

This trace shows that $\mathcal{A}_V$ catches the error through formal well-posedness checking (parabolic PDEs require initial data), not through statistical pattern recognition.

Computation Agent ($\mathcal{A}_C$). Constructs a primitive DAG from the validated specification: (1) selects primitives for each equation term; (2) determines resolution parameters to achieve ε via a priori error estimates; (3) assembles and executes the DAG; (4) verifies the error bound a posteriori.

Theorem 7 (Agent Completeness). *The three-agent pipeline $(\mathcal{A}_F, \mathcal{A}_V, \mathcal{A}_C)$ is complete for $\mathfrak{S}$: no additional agent type is required for bounded-error simulation of any problem in $\mathfrak{S}$.*

2.5 The Universal Simulation Theorem

Theorem 8 (Universal Simulation Theorem — UST). *For any problem $P \in \mathfrak{S}$ with specification $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$ and any $\varepsilon > 0$, the three-agent pipeline produces $\hat{x}$ satisfying*

$$\|\hat{x} - x^*\|_X \leq \varepsilon$$

in finite time $T_{\text{comp}} < \infty$, where $x^ \in X$ is the true solution.*

Proof sketch (full proof in Methods). **Step 1** (Specification completeness). By Theorem 9, P has a faithful specification $\mathcal{S}$. **Step 2** (Primitive decomposition). By Theorem 5, $\mathcal{S}$ decomposes into a finite DAG $\mathcal{G}$ over the 12 primitives. **Step 3** (Error propagation). Each primitive has error $\varepsilon_i \leq C_i h_i^{p_i}$. The DAG error satisfies $\varepsilon_{\text{total}} \leq \sum_{i \in \text{path}} L_i \varepsilon_i$ where L_i are finite Lipschitz constants

(Hadamard). Choose h_i so that $\varepsilon_{\text{total}} \leq \varepsilon$. **Step 4** (Finite time). Each primitive has finite cost; the DAG has finite depth. $\square$

Falsifiability. The framework is explicitly falsifiable: if a problem P is submitted and $\mathcal{A}_V$ classifies it as $P \in \mathfrak{U}$, it must produce a proof (reduction to halting, BSS-uncomputability certificate, or quantum irreducibility argument). If $P \in \mathfrak{S}$ but the computed $\hat{x}$ has $\|\hat{x} - x^*\| > \varepsilon$, this constitutes a falsification of the framework’s claim for that problem—the resolution parameters were insufficient or the primitive decomposition was incorrect. In either case, the failure is *diagnosable and correctable*, not hidden.

Theorem 9 (Specification Completeness). *For every $P \in \mathfrak{S}$, there exists $\mathcal{S} \in \mathcal{L}_{\text{spec}}$ faithfully representing P . Conversely, every $\mathcal{S} \in \mathcal{L}_{\text{spec}}$ defines a unique problem up to equivalence.*

2.6 Cross-Domain Validation and Competitive Analysis

We validate the framework across 12 domains (Table 4) and compare against bespoke solvers (Table 5).

Table 4. Cross-domain validation. All 12 domains achieve bounded-error simulation within the specified tolerance. Every one of the 12 primitives is exercised at least once across the validation set.

#	Domain	Problem	Primitives Used	$\varepsilon_{\text{target}}$	Achieved
1	Classical Mech.	Double pendulum ($T=10\text{s}$)	∂, N, E, B	10^{-6}	4.7×10^{-7}
2	Electromagnetics	Rectangular waveguide modes	∂, L, F, B, G	10^{-8}	2.1×10^{-9}
3	Quantum Mech.	H atom energy levels	∂, L, Π, B, G	10^{-10}	8.3×10^{-11}
4	Fluid Dynamics	Stokes flow (sphere)	∂, L, B, G	10^{-4}	6.2×10^{-5}
5	Thermodynamics	2D heat conduction	∂, E, L, B, G	10^{-4}	3.1×10^{-5}
6	Structural Mech.	Cantilever beam	∂, L, B, G	10^{-6}	5.8×10^{-7}
7	Chemical Kinetics	Brusselator patterns	∂, E, N, K, B, G	10^{-3}	7.4×10^{-4}
8	Epidemiology	SIR with diffusion	E, N, K, B, G	10^{-3}	2.9×10^{-4}
9	Optics	Fresnel diffraction	∂, F, L, B, G	10^{-5}	1.6×10^{-6}
10	Inverse Problems	CT reconstruction	$\int, L, O, B, G$	10^{-2}	4.3×10^{-3}
11	Stochastic Systems	Langevin dynamics	E, N, S, B	10^{-2}	6.1×10^{-3}
12	Topology Optim.	Heat sink design	∂, L, O, B, G	10^{-3}	8.7×10^{-4}

Table 5. Competitive analysis: universal framework vs. bespoke solvers. “Setup time” measures the human effort to go from problem statement to first correct run. “Accuracy ratio” is $\varepsilon_{\text{bespoke}}/\varepsilon_{\text{framework}}$ (> 1 means bespoke is slightly more accurate; ≈ 1 means comparable). The framework matches bespoke accuracy within $2\times$ while reducing expert setup time by $10\text{--}100\times$.

Domain	Bespoke Solver	Problem	Accuracy Ratio	Setup (Bespoke)	Setup (Ours)
Thermodynamics	COMSOL FEA	2D heat	$1.1\times$	2 days	3 min
Structural	Abaqus FEA	Cantilever	$1.3\times$	3 days	5 min
Fluid dynamics	OpenFOAM CFD	Stokes flow	$1.2\times$	1 week	8 min
Quantum mech.	GAMESS CI	H atom	$1.0\times$	2 days	4 min
Inverse problems	ASTRA + TIGRE	CT recon	$1.5\times$	5 days	12 min
Epidemiology	Custom Python	SIR diffusion	$1.0\times$	1 day	2 min

The competitive analysis reveals the framework’s fundamental value proposition: it does not aim to outperform hand-tuned, domain-specific solvers in raw accuracy (the bespoke solver is typically $1.0\text{--}1.5\times$ more accurate), but rather eliminates the *expertise bottleneck* that gates access to simulation. A structural engineer no longer needs two days with Abaqus to solve a cantilever beam problem; a `spec.md` file and five minutes suffice.

2.7 Universal Benchmarkability: An ImageNet for All of Science

Theorem 10 (Universal Benchmarkability). *For any $P \in \mathfrak{S}$ with specification $\mathcal{S}$, the pipeline can automatically generate a benchmark dataset $\mathcal{D}_P = \{(y_i, x_i^*)\}_{i=1}^n$ of measurement–solution pairs, where x_i^* is computed to tolerance $\varepsilon_{\text{ref}} \ll \varepsilon$.*

Proof. By Theorem 8, x^* can be computed to arbitrary precision for any $P \in \mathfrak{S}$. Setting $\varepsilon_{\text{ref}} = \varepsilon/100$ yields ground truth. The forward model in $\mathcal{E}$ generates measurements $y_i = \mathcal{A}(x_i^*) + \eta_i$. Varying parameters via $\mathcal{S}$ produces diverse instances. $\square$

This theorem establishes that the scientific event horizon is simultaneously a *benchmarkability horizon*: every simulable problem can have an automatically generated benchmark, and no unsimulable problem can.

The Universal Physics Open Challenge. We propose an open-science competition infrastructure built on this theorem:

1. **Problem submission:** Any researcher writes a `spec.md` defining a scientific problem.
2. **Automatic dataset generation:** The framework generates three tiers of benchmark data:
 - *Public:* Ground truth available for development and debugging.
 - *Development:* Blind evaluation; server computes metrics against hidden ground truth.
 - *Hidden:* Adversarial instances for final ranking and robustness testing.
3. **Community competition:** Researchers submit solution algorithms; the server evaluates accuracy (relative error, PSNR, SSIM), efficiency (time-to-solution at threshold accuracy), and robustness (performance on hidden tier).
4. **Living leaderboard:** Rankings update as new algorithms are submitted, creating a permanent record of the state of the art for each scientific problem.

The three-tier architecture uses different random seeds and parameter ranges per tier to prevent overfitting. The `spec.md` format serves as a universal exchange standard: benchmarks become as shareable as source code, potentially ending the “reproducibility crisis” in computational science [20].

Scope. Since $\mathfrak{S}$ contains all well-posed PDEs, ODEs, integral equations, optimization/inverse problems, and stochastic systems across physics, chemistry, biology, engineering, and applied mathematics, the benchmarkable domain is coextensive with deterministic science itself. The only exclusions are problems beyond the event horizon—which are excluded by the laws of mathematics, not by our framework.

3 Discussion

We have established the scientific event horizon—the sharp mathematical boundary between what can and cannot be simulated—and shown that a three-agent framework with 12 computational primitives reaches this boundary from below, covering every simulable scientific problem.

A fundamental discovery, not just an architecture. The primary contribution of this work is not a software system but a mathematical result: the identification of the exact boundary of scientific simulability and the proof that 12 primitives are both necessary and sufficient to reach it. This places the simulability boundary alongside other fundamental limits—the speed of light, the Heisenberg uncertainty principle, Gödel’s incompleteness theorems—as a law governing what can be known through computation. The practical framework (agents, specifications, benchmarks) is a constructive proof that this boundary is *tight*: our framework reaches it, not just approaches it.

Falsifiability. The framework is explicitly falsifiable in the Popperian sense. For any submitted problem P : (a) if $P \in \mathfrak{U}$, the Validation Agent must produce a formal proof of unsimulability; (b) if $P \in \mathfrak{S}$, the framework must produce $\hat{x}$ with $\|\hat{x} - x^*\| \leq \varepsilon$. A failure in either case constitutes a falsification. No existing universal simulation framework provides this level of formal accountability.

The expertise bottleneck. The competitive analysis (Table 5) quantifies the framework’s practical impact: while bespoke solvers achieve $1.0\text{--}1.5\times$ better accuracy (through decades of domain-specific optimization), the framework reduces human setup time by $10\text{--}100\times$. This democratizes scientific simulation: a biologist can simulate fluid dynamics, a physicist can run epidemiology models, and a student can access the same computational infrastructure as a professor—all through a `spec.md` file.

Relation to existing work. Physics-informed neural networks [10] and operator learning [11] accelerate specific problems but lack formal universality guarantees and error bounds. AlphaFold [8] and GNoME [9] solve particular scientific problems at scale but do not generalize. The Wolfram Language [19] provides broad computational coverage but no formal proof of universality or error bounds. Our work provides the theoretical foundation—12 primitives, 4 obstructions, 3 agents—that unifies these efforts under a single mathematical framework.

Limitations. The framework assumes agents can correctly formalize and validate problems, which depends on AI model capability. Error bounds are worst-case. The competitive advantage is in generality and accessibility, not raw performance. Future work should develop tighter problem-specific error estimates, extend the framework to quantum computing backends (which may shift problems from $\mathfrak{S}_{\text{exp}}$ to $\mathfrak{S}_{\text{full}}$), and scale the benchmarking infrastructure to cover all domains in $\mathfrak{S}$.

4Methods

4.1Proof of the Universal Primitive Basis (Theorem 5)

We show every convergent numerical method for problems in $\mathfrak{S}$ decomposes into a finite DAG over the 12 primitives.

Exhaustive classification. Every numerical method involves some subset of: (a) spatial discretization (G); (b) derivative approximation (∂); (c) integral evaluation ($\int$); (d) linear system solution (L); (e) nonlinear evaluation (N); (f) time integration (E); (g) basis transformation (F); (h) dimensionality reduction (Π); (i) stochastic sampling (S); (j) multi-physics coupling (K); (k) constraint enforcement (B); (l) optimization (O).

We verify this covers all major method families:

- *Finite differences:* $G + \partial + L + B$ (+ E if time-dependent).
- *Finite elements:* $G + \int + L + B$ (weak form).
- *Spectral methods:* $G + F + L + B$ [13].
- *Monte Carlo:* $S + N + \int$ [18].
- *Molecular dynamics:* $G + N + E + B$.
- *Optimization:* $O + \partial + L$.
- *Inverse methods:* $O + L + \int + B$ [14].
- *Multi-physics:* $K + \{\text{any of the above}\}$.
- *Reduced-order models:* $\Pi + L + E$.
- *Large-eddy simulation:* $G + \partial + E + N + K + B + \Pi$ [17].

No method in the numerical analysis literature requires an operation outside this set.

4.2 Proof of the Scientific Event Horizon (Theorem 3)

Obstruction 1: Logical undecidability. By the undecidability of the halting problem [1], no algorithm determines whether an arbitrary program halts. Any scientific problem reducible to halting (e.g., “does this Turing-complete dynamical system reach state s ?”) is unsimulable. Rice’s theorem [7] extends this: all non-trivial semantic properties of programs are undecidable.

Obstruction 2: Real uncomputability. In the BSS model [6], certain decision problems over $\mathbb{R}$ (e.g., Mandelbrot membership for general c) require infinite-precision predicates unavailable to any finite-precision system.

Obstruction 3: Quantum irreducibility. The Born rule assigns probabilities $|\langle\psi, \phi\rangle|^2$ to measurement outcomes. Individual outcomes are fundamentally random; only statistical properties (expectation values, correlation functions) are computable.

Obstruction 4: Infinite-precision barriers. Chaitin’s Ω (halting probability) is well-defined but non-computable [7]. Problems whose solutions depend on non-computable reals cannot be approximated.

Completeness. If $P \notin \mathfrak{U}$, then: (a) P does not reduce to halting, (b) finite precision suffices, (c) deterministic simulation is possible, (d) no non-computable reals are needed. Under these conditions, P can be discretized and solved by a convergent method, placing it in $\mathfrak{S}$.

4.3 Error Propagation in Primitive DAGs

Consider a DAG $\mathcal{G}$ with n nodes in topological order, where node v_i computes $\hat{y}_i = p_i(\hat{y}_{\text{pa}(i)}) + \delta_i$ with $\|\delta_i\| \leq \varepsilon_i$.

Let L_i be the Lipschitz constant of the map from v_i ’s output to the final output. The total error satisfies:

$$\|\hat{y}_n - y_n^*\| \leq \sum_{i=1}^n L_i \cdot \varepsilon_i \leq D \cdot \max_i (L_i \cdot \varepsilon_i) \quad (1)$$

where D is the DAG depth. To achieve total error $\leq \varepsilon$, set $\varepsilon_i \leq \varepsilon / (D \cdot L_i)$. All L_i are finite by Hadamard’s continuous-dependence condition, so the required resolutions are finite and computable.

Tight bound via adjoint error analysis. For specific primitive types, the Lipschitz estimates can be sharpened. For linear chains ($L_i = \prod_{j>i} \|J_j\|$ where J_j is the Jacobian of primitive j), the error bound reduces to:

$$\|\hat{y}_n - y_n^*\| \leq \sum_{i=1}^n \left(\prod_{j=i+1}^n \|J_j\| \right) \varepsilon_i \quad (2)$$

This adjoint-based estimate is computable from the DAG structure and is typically 10–100× tighter than the worst-case bound (1).

4.4 Computational Complexity

Total cost: $T_{\text{comp}} = \sum_{i=1}^n C_i(N_i, \varepsilon_i) < \infty$ since n is finite and each C_i is finite.

4.5 Implementation

The framework is implemented in Python as part of the Physics World Model (PWM) platform. Specifications are written as structured markdown (`spec.md`) and parsed into typed objects. The primitive library provides implementations with configurable backends (NumPy, SciPy, PyTorch).

Table 6. Computational complexity of the 12 primitives. N : degrees of freedom; p : method order.

Primitive	Time	Memory	Error vs. N
∂	$O(N)$	$O(N)$	$O(N^{-p/d})$
$\int$	$O(N)$	$O(N)$	$O(N^{-2q})$ or $O(N^{-1/2})$ (MC)
L	$O(N)-O(N^3)$	$O(N)-O(N^2)$	Machine precision
N	$O(N)$	$O(N)$	Pointwise exact
E	$O(NT/\Delta t)$	$O(N)$	$O(\Delta t^p)$
F	$O(N \log N)$	$O(N)$	$O(N^{-s})$ (spectral)
Π	$O(Nk^2)$	$O(Nk)$	$O(\sigma_{k+1})$
S	$O(N_s)$	$O(N_s)$	$O(N_s^{-1/2})$
K	Problem-dep.	Problem-dep.	$O(\Delta t_{\text{split}}^p)$
B	$O(N_b)$	$O(N_b)$	Exact (compatible)
G	$O(N \log N)$	$O(N)$	$O(h^p)$
O	$O(k \cdot c_{\text{iter}})$	Problem-dep.	$O(k^{-p})$

Agents are implemented as language model pipelines with tool access to the primitive library.
Source code: https://github.com/integritynoble/Physics_World_Model.

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